CHAPTER 1 - INTRODUCTION

1.1 Background of the Study

Work-Integrated Learning (WIL) is an important part of many occupational and technical qualifications because it gives students an opportunity to gain practical workplace experience and apply the knowledge and skills they have learned during their studies. During a WIL placement, students are required to complete workplace activities, record the work they have done, complete daily documentation and provide evidence of their activities.

Currently, the management of WIL activities can involve students completing documents daily and sending them to their workplace mentors or supervisors for verification and signing. Students then must wait for the signed documents to be returned before they can continue with the submission or verification process. These documents may be exchanged through email or other communication methods, which can make it difficult for students and academic staff to keep track of all required documents.

When documents are stored in different locations or sent between students and mentors, there is also a risk that documents may be misplaced, accidentally deleted or not returned on time. Coordinators and lecturers may also have difficulty determining which students have completed their daily requirements and which students are still waiting for documentation to be completed.

TrackWIL is proposed as a digital solution to improve this process. The system will provide students with a central platform where they can complete their daily WIL activities, submit their documentation and upload evidence. Workplace mentors will be able to access and verify the relevant submissions, while coordinators and lecturers will be able to monitor student progress.

1.2 Problem Statement

The current WIL documentation process can be time-consuming because students are required to complete daily documents and send them to their workplace mentors for verification and signing. Students then must wait for the signed documents to be returned. When this process is managed through emails, separate files or other manual methods, documents can be misplaced, accidentally deleted or returned late.

This affects students, because they need to keep track of their daily documents and ensure that they are completed and signed by their mentors. Delays in receiving signed documents can also make it difficult for students to keep their WIL records up to date.

It also affects WIL coordinators, who need to monitor whether students are completing their required activities and receiving the necessary verification. Checking documents from multiple students can become difficult and time-consuming.

Lecturers are also affected because they need to identify which students are up to date and which students are falling behind. Without a centralised system, it can be difficult to obtain an accurate and current view of each student's WIL progress.

1.3 Research Aim

The aim of this project is to design and develop TrackWIL, a centralized digital WIL management application that allows students to record their daily workplace activities, submit documentation and evidence for mentor verification, and enables coordinators and lecturers to monitor student progress.

1.4 Research Objectives

- Analyse the current WIL documentation and monitoring process to identify the challenges experienced by students, workplace mentors, coordinators and lecturers.
- Design an integrated WIL management system that allows students to submit daily WIL documents and evidence to their workplace mentors for verification.
- Develop the TrackWIL application with separate dashboards for students, mentors, coordinators and lecturers.
- Implement features that allow users to submit, verify, store and monitor WIL documents and daily activities digitally.
- Test and evaluate the TrackWIL application to determine whether it improves the efficiency of WIL documentation, verification and progress monitoring.

1.5 Research Questions

1. What challenges are experienced by students, workplace mentors, coordinators and lecturers when managing WIL documentation and activities using the current process?
2. How can an integrated digital system improve the submission and verification of students' daily WIL documents?
3. How can TrackWIL support students, mentors, coordinators and lecturers through role-specific dashboards?
4. How can digital submission, verification, storage and monitoring of WIL documents improve the management of student WIL activities?
5. To what extent can TrackWIL improve the efficiency of WIL documentation, verification and student progress monitoring?

1.6 Proposed Software Solution

The proposed software solution is TrackWIL, a web-based application designed to improve the management of Work-Integrated Learning activities and documentation.

The system will provide different dashboards based on the user's role. Students will have a dashboard where they can record their daily WIL activities, complete their required documents and upload evidence of the work they have completed. They will then be able to submit their documents through the system to their workplace mentor for verification.

Workplace mentors will have access to the relevant student submissions. They will be able to review the students' daily documentation, verify the information and approve the required documents. The student will then be able to see whether the document has been approved, rejected or is still waiting for the mentor's response.

WIL coordinators will have a dashboard where they can monitor students and see whether their daily documentation has been submitted and verified. They will also be able to identify outstanding documents and students who may require attention.

Lecturers will have a monitoring dashboard that provides an overview of student progress. They will be able to see which students are up to date, which students have outstanding work, and which students may be falling behind.

1.7 Scope

The TrackWIL project will include the design and development of a web-based WIL management system with the following functionality:

Student functionality

- Student registration and login.
- Student dashboard.
- Recording of daily WIL activities.
- Completion of daily WIL documentation.
- Uploading of work and supporting evidence.
- Submission of documents to workplace mentors.
- Viewing the status of submitted documents.
- Viewing mentor feedback or verification.
- Monitoring personal WIL progress.

Mentor functionality

- Mentor login.

- Viewing assigned students.
- Viewing student daily submissions.
- Reviewing uploaded documents and evidence.
- Approving or rejecting submissions.
- Verifying required documents.
- Providing feedback to students.

Coordinator functionality

- Coordinator login.
- Viewing registered WIL students.
- Monitoring daily submissions.
- Checking whether documents have been verified.
- Identifying missing or outstanding submissions.
- Monitoring student WIL progress.

Lecturer functionality

- Lecturer login.
- Viewing student progress.
- Identifying students who are up to date or behind.
- Viewing outstanding WIL activities and documentation.
- Monitoring student performance and progress.

System functionality

- User authentication.
- Role-based access.
- Document upload and storage.
- Submission tracking.

- Verification/status tracking.
- Progress monitoring.
- Basic reporting.

1.8 Significance of the Study

The TrackWIL project is expected to provide benefits to several groups involved in the WIL process.

Students

TrackWIL will make it easier for students to manage their daily WIL documentation. Instead of relying mainly on sending documents through email and waiting for them to be returned, students will have one central place to submit their documents and check their verification status.

Students will also be able to see which documents have been completed and which are still outstanding. This can help students stay organized and keep track of their WIL requirements.

Workplace Mentors

Mentors will have easier access to the documents submitted by students assigned to them. They will be able to review, verify and sign documents through the system rather than relying on separate communication channels.

This can make the verification process more organized and provide a record of completed mentor approvals.

WIL Coordinators

Coordinators will have a centralized view of student submissions and verification statuses. Instead of manually checking different documents or asking students for updates, coordinators can use the dashboard to identify missing or outstanding work.

This can reduce the administrative workload and make monitoring more efficient.

Lecturers

Lecturers will be able to monitor student progress from one dashboard. They can identify students who are up to date and students who are falling behind with their WIL requirements.

This can allow lecturers to identify problems earlier and provide support where necessary.

Educational Institutions

TrackWIL can help educational institutions improve the organization and management of WIL documentation. Centralizing information can reduce dependence on emails, spreadsheets and separate documents and provide a clearer record of student WIL progress.

Software Engineering Field

The project demonstrates how software engineering can be used to solve a real-world problem in education and workplace training. It will apply concepts such as requirements analysis, database design, user authentication, role-based access, document management, system development and software testing.

1.9 Chapter Summary

This chapter introduced the TrackWIL project and discussed the challenges associated with the current management of Work-Integrated Learning documentation and activities. Students currently need to complete their daily WIL documents, submit them to workplace mentors for verification or signing and wait for the completed documents to be returned. This process can make it difficult to track submissions and identify outstanding documentation.

The chapter presents the research aim, objectives and research questions that will guide the development of TrackWIL. The proposed solution is a centralized web-based application that will allow students to submit daily WIL documents and evidence, while workplace mentors will be able to review and verify these submissions. Coordinators and lecturers will be able to monitor student submissions and overall progress through their respective dashboards.

The scope of the project was also defined by identifying the features that will be included and excluded from the system. Finally, the significance of the study explained how TrackWIL can benefit students, workplace mentors, WIL coordinators, lecturers and educational institutions.